Spaced Repetition Prompts

Definitions, Theorems, and Proof Moves

Engineering Math to Pure Math

May 2026

1 Schedule

Review each card on days 1, 3, 7, 14, and 30. If you miss it, reset the card. The point is not shame; the point is retrieval.

2 Cards

1. What is a mathematical statement?
2. What is a predicate?
3. Negate: $\forall x \in A,\ P(x)$.
4. Negate: $\exists x \in A,\ P(x)$.
5. What does $P \Rightarrow Q$ mean?
6. State the contrapositive of $P \Rightarrow Q$.
7. What must you prove to show $A \subseteq B$?
8. What must you prove to show $A = B$?
9. Define $A \cup B$.
10. Define $A \cap B$.
11. Define $A \setminus B$.
12. Define $\mathcal{P}(A)$.
13. Define a function $f : A \to B$.
14. What is the difference between codomain and image?
15. Define injective.
16. Define surjective.
17. Define bijective.
18. What is $f^{-1}(S)$ for $S \subseteq B$?
19. Define an equivalence relation.
20. What is an equivalence class?
21. Define a partial order.
22. State the induction principle.
23. What changes in strong induction?
24. What is a counterexample?
25. What is a witness in an existence proof?
26. Prove in one sentence that the sum of two even integers is even.
27. Why is $\sqrt{2}$ irrational?
28. What does countable mean?
29. Why is $\mathbb{Q}$ countable?
30. What is the core idea of Cantor diagonalization?
31. Define convergence of a sequence.
32. Negate the definition of sequence convergence.
33. Define continuity at a point.
34. What is an open ball?
35. Define an open set in a metric space.

36. Define a group.
37. Define a subgroup.
38. What is a group homomorphism?
39. What is a kernel?
40. Why are kernels related to quotients?
41. Define a ring.
42. Define an ideal.
43. What is a topology?
44. What does compactness try to capture?
45. What is a manifold, informally?
46. What is a chart?
47. What is an affine variety?
48. What is a coordinate ring?

3 Long-Form Recall Prompts

1. Explain the difference between solving an equation and proving a theorem.
2. Explain why definitions in pure mathematics feel stricter than formulas in engineering mathematics.
3. Tell the story of how calculus leads to real analysis.
4. Tell the story of how modular arithmetic leads toward abstract algebra.
5. Explain why topology can talk about continuity without measuring distance.
6. Explain how polynomial equations can define geometric shapes.
7. Describe one idea that appears in analysis, algebra, and topology under different names.